

Week 5 explained (easy words, but nothing skipped)

Think of Week 5 like this story:

You're building an app. If you build it only by staring at tech (features, APIs, "wow AI"), you can still fail—because real users are messy. So HCI says: **design it in stages**, and **evaluate it properly**, using **rules/heuristics** so you don't miss obvious usability problems.

1) The Design Process (Design stages)

A) Technology Myopia (the “tunnel vision” warning)

Technology myopia = when you focus too much on “the technology” and ignore the human/social context of use.

To avoid tech myopia, designs must be sensitive to:

1. **Human-human communication**: Many systems are really about people communicating through tech (WhatsApp, Bykea driver-customer chat, comments, reviews). If the design makes communication awkward, the system fails socially.
2. **Implicit (tacit) knowledge**: People do many things without explicitly thinking (workarounds, habits). A good system fits these unspoken practices instead of forcing users to “explain their whole brain.”
3. **Non-technical aspects of work**: Real environments have emotions, politics, shortcuts, stress, urgency. A “perfect” technical flow can still fail in real life.

What to memorize (teacher nitpick zone):

- Definition of **technology myopia** + the **3 sensitivities** (human-human, implicit knowledge, non-technical aspects).
-

B) Conceptual Design (the “idea + structure” stage)

Conceptual design is the **high-level planning** of the system: what it is, how information is organized, and what solution approach you're choosing.

It involves:

- **Structuring the information space**: organizing content/features so users can find things logically.
 - Example: In a grocery app, do you structure by “Categories” (Dairy, Snacks) or by “Deals” vs “Regular” vs “Past orders”?

- **Creating alternative solutions:** brainstorming multiple ways to solve the same user problem.
 - Example: For booking a ride, options could be: a single screen flow, a step-by-step wizard, or a map-first flow.
- **Choosing which concept to pursue:** selecting the best concept based on user needs + project goals.

What to memorize:

- The phrase “**structuring information space, creating alternatives, selecting the concept**”.
-

C) Physical Design (the “actual UI” stage)

Physical design = turning the concept into **concrete screens and UI pieces**.

It answers:

- **What it looks like:** colors, typography, layout style, branding
- **What components it needs:** buttons, forms, menus, navigation bars, etc.
- **How screens are laid out:** exact screen-by-screen layout and flow

Mini example (Yango task):

Conceptual design: “User books ride + selects payment + sees traffic.”

Physical design: where is “Comfort” option shown? how many taps? where is “Traffic display”? is it a toggle on map? is it hidden in settings?

What to memorize:

- Physical design is about **look + components + screen layout**.
-

D) Standards and Tools (how we keep consistency)

Standards/tools help keep design consistent and efficient. In your slides:

- **Graphical libraries / UI toolkits**
- **Visual interface builders**
- **Web development tools**

And your notes mention that toolkits enforce consistency via reusable components.

Easy way to understand:

If every designer makes their own buttons from scratch, your app becomes messy. Standards/toolkits make buttons, spacing, and behavior consistent.

What to memorize:

- Examples list: graphical libraries, UI toolkits, visual builders, web tools.
-

2) Evaluation and Measurement (How we check if the design is good)

A) Traditional measurements (classic usability metrics)

The “classic trifecta” you must know:

1. **Task performance (effectiveness)**: can users complete tasks? (success rate)
2. **Time performance (efficiency)**: how long do tasks take?
3. **User satisfaction**: how users feel (often questionnaires like SUS)

Example: Booking a ride:

- effectiveness = did they successfully book?
- efficiency = time taken to book
- satisfaction = rating (“easy to use?”)

Memorize exactly: task performance + time performance + satisfaction.

B) Why traditional measurements are not enough (modern UX is deeper)

Your lecture says traditional measures **don’t capture**:

- **Why people abandon an interface**
- **Discretionary use for enjoyment** (YouTube/games)
- **Emotion and trust**

So we need **new forms of measurement** like:

- feeling of community
- enjoyment / emotional engagement
- satisfaction from achievements (gaming/virtual worlds)
- physiological measures like **EEG/EMG** (objective signals)

Easy example:

Two apps can have the same “task completion time,” but one feels stressful and untrustworthy, so users quietly stop using it. That’s why “emotion/trust” matter.

What to memorize:

- “Traditional metrics miss emotion + trust + enjoyment + abandonment.”
 - Examples of “new measurements”: community, enjoyment, EEG/EMG.
-

3) Triangulation (the research reliability idea)

Triangulation = “don’t trust one method only”

No data method is perfect, so triangulation means using **multiple researchers and/or multiple methods** to study the same phenomenon.

Key message: **One paper ≠ scientific truth**. Confidence grows when different methods/researchers reach the same conclusion.

Example: Evaluating Yango booking flow

- Method 1: usability test (watch users struggle)
 - Method 2: survey (many users report confusion)
 - Method 3: analytics (drop-off at payment screen)
- If all point to “payment step is confusing,” that’s strong evidence.

What to memorize:

- Definition + “one paper ≠ truth” + “multiple methods/researchers.”
-

4) Types of research (Descriptive vs Relational vs Experimental)

Your notes table says:

1. **Descriptive**: describes what is happening (“X is happening”).
 - Methods: observations, field studies, focus groups, interviews.
2. **Relational**: finds relationships (“X is related to Y”).
 - Methods: observations, field studies, surveys.
3. **Experimental**: finds cause (“X is responsible for Y”).

- Methods: controlled experiments.

Examples:

- Descriptive: “Users often enable traffic display while booking rides.”
- Relational: “Users who use Easypaisa report higher satisfaction.”
- Experimental: “Adding a payment summary screen causes fewer payment errors.”

Memorize: the **claim style** (X is happening / related / responsible) + typical methods.

5) Experimental design (how experiments are set up)

Your notes emphasize experiments are best for **cause-and-effect** and hinge on things like **random assignment**.

Your slide also highlights key planning questions: how many **independent variables** and how many values/levels each has.

Independent variable (IV) = what you change (e.g., “payment selection UI: dropdown vs cards”).

Dependent variable (DV) = what you measure (time, errors, satisfaction).

What to memorize:

- Experiments study **cause-effect**, involve **independent variables and levels**.
-

6) Other data collection methods (when you don't do experiments)

From your Week 5 lecture:

A) Surveys

Good for: lots of responses, fast, geographically dispersed, “big picture,” and can estimate population if sampling is good.

B) Diaries (ethnography-related)

A diary is written by the user **as events occur**, good for things changing over time (like mood). Useful when after-the-fact recall is biased/inaccurate.

C) Case studies

In-depth study of one or few instances in real context; time/labor intensive but informative and helps generate hypotheses.

D) Interviews & focus groups

Directly ask users; focus groups add group discussion dynamics.

E) Ethnography

Observe people in real context, understand behavior/culture and “what they really do,” not what they claim.

Memorize: definitions and “best use case” (survey = big picture, diary = over time + recall bias, case study = deep context).

7) Design rules: Standards vs Guidelines vs Principles

Goal of interaction design

Goal = **designing for maximum usability.**

Two dimensions to classify design rules

Rules vary by:

1. **Authority:** must-follow vs suggested
2. **Generality:** broad applicability vs specific context

This creates:

- **Standards:** high authority, low generality (must follow, specific)
- **Guidelines:** lower authority, high generality (recommended good practice)
- **Principles:** abstract, lower authority, high generality (fundamental ideas)

Easy example:

- Standard: “Use iOS back gesture behavior exactly.”
- Guideline: “Provide informative feedback.”
- Principle: “Support the user’s mental model.”

Memorize exactly: authority + generality dimensions; and where standards/guidelines/principles fall.

8) Heuristics and Golden Rules (practical checklists)

Heuristics = “broad brush” rules, used like a checklist for good design.
Common collections include Shneiderman, Norman, Nielsen.

A) Shneiderman’s 8 Golden Rules (must memorize)

These are explicitly listed and explained in your notes.

1. **Strive for consistency** (visual + behavior).
 - Example: “Book” button looks and behaves similarly across ride, courier, truck.
2. **Cater to universal usability** (novice + expert; shortcuts for frequent users).
3. **Offer informative feedback** (system responds clearly).
 - Example: progress indicator when searching for drivers.
4. **Design dialogs to yield closure** (clear end to tasks).
 - Example: “Ride confirmed” screen with summary.
5. **Error prevention + simple handling** (prevent first; clear recovery).
6. **Permit easy reversal of actions** (undo/redo).
 - Example: cancel booking within a grace period.
7. **Support internal locus of control** (user feels in control).
8. **Reduce short-term memory load** (recognition > recall, chunking).
 - Example: show saved addresses and recent destinations so users don’t remember exact spellings.

Memorize: all 8 names + what each means in one line + 1 example each.

B) Nielsen’s 10 Usability Heuristics (must memorize)

Your notes list them; these are extremely testable in HCI.

1. **Visibility of system status** (always keep users informed)
2. **Match with the real world** (user language, real-world conventions)
3. **User control and freedom** (undo, cancel, exits)
4. **Consistency and standards**
5. **Error prevention**
6. **Recognition rather than recall**
7. **Flexibility and efficiency of use** (shortcuts, customization)
8. **Aesthetic and minimalist design**

9. **Help users recognize/diagnose/recover from errors**
10. **Help and documentation**

How teachers nitpick: they'll give you a UI example and ask: *which heuristic is violated?*

Example: If the "Apply Rs 100 voucher" option is hidden and users can't find it, that's often **recognition rather than recall + visibility of system status** (if voucher applied isn't shown) + sometimes **aesthetic/minimalist** if clutter hides key controls.

9) How the class exercise connects (Yango + Heuristics)

Your "Nielsen Heuristic Form" exercise makes you perform realistic tasks in Yango: booking ride, courier, truck, grocery items, enabling traffic display, selecting JazzCash/Easypaisa, applying voucher, scheduling times, instructions, etc.

This exercise is basically:

- do tasks (like a usability walkthrough)
- then evaluate the screens using heuristics (Nielsen / Shneiderman)

Example mapping (quick):

- If enabling traffic display is hidden: violates **visibility of system status** and **recognition rather than recall**.
 - If you can't undo a wrong pickup time: violates **user control and freedom** / "easy reversal".
 - If voucher applies but you don't get a clear confirmation: violates **informative feedback** and **closure**.
-

What to memorize (Week 5 "exam bullets")

If you memorize only one section perfectly, do **this list**:

1. **Technology myopia** definition + 3 sensitivities.
2. **Conceptual vs physical design** (what each includes).
3. Traditional usability metrics: **effectiveness, efficiency, satisfaction**.
4. Why traditional metrics are insufficient + examples of new measurements (emotion/trust/community/EEG-EMG).
5. **Triangulation** meaning + why "one study isn't truth."
6. Research types: descriptive/relational/experimental (claim + methods).

7. Design rules classification: **authority + generality**, and **standards/guidelines/principles**.
 8. **Shneiderman's 8** (names + meaning).
 9. **Nielsen's 10** (names + identify violations in examples).
-

Practice Questions (with answers + explanation)

A) 5 MCQs (each with answer + why)

MCQ 1

Technology myopia mainly means:

- A) Using too many colors in UI
- B) Focusing on technology while ignoring human/social context
- C) Making the interface too minimal
- D) Using surveys instead of experiments

Answer: B. Because technology myopia is specifically the “tunnel vision” focus on tech, ignoring people/context.

MCQ 2

Which is NOT part of conceptual design?

- A) Structuring the information space
- B) Creating alternative solutions
- C) Choosing typography and color palette
- D) Deciding which concept to pursue

Answer: C. Typography/colors are part of **physical design** (visual details).

MCQ 3

The traditional usability metric trio is:

- A) Trust, emotion, community
- B) Learnability, memorability, aesthetics
- C) Task performance, time performance, user satisfaction
- D) Surveys, interviews, case studies

Answer: C. That's the classic trifecta.

MCQ 4

Triangulation is best described as:

- A) Using one perfect method to avoid bias
- B) Using multiple methods/researchers to study the same phenomenon
- C) Only using experiments for evaluation
- D) Repeating the same survey twice

Answer: B. Triangulation is specifically multiple methods and/or multiple researchers.

MCQ 5

“Standards” are:

- A) Low authority, high generality
- B) High authority, low generality
- C) Low authority, low generality
- D) High authority, high generality

Answer: B. Standards must be followed and are usually specific to a platform/org.

B) 5 Short Answer Questions (with answers + explanation)

SAQ 1: Define technology myopia and list the three things designs must be sensitive to.

Answer: Technology myopia is focusing too narrowly on technology and ignoring the human/social context. The design must be sensitive to **human-human communication**, **implicit knowledge**, and **non-technical aspects of work**.

Explanation: This is tested because it explains why “good tech” can still fail in real life.

SAQ 2: What is the difference between conceptual design and physical design?

Answer: Conceptual design is the high-level structure and deciding the design concept (organize info, alternatives, selection). Physical design is the concrete UI: look/feel, components, and screen layouts.

Explanation: Conceptual = “what and how it’s organized.” Physical = “what users actually see and click.”

SAQ 3: Name the three traditional usability measurements and one limitation.

Answer: Task performance (effectiveness), time performance (efficiency), user satisfaction. A limitation is they don’t capture things like **emotion and trust** or why users abandon the interface.

Explanation: Modern products succeed/fail on trust and feelings, not just speed.

SAQ 4: Explain triangulation in one or two lines and why it matters.

Answer: Triangulation means using multiple methods and/or researchers to study the same thing; it matters because no single method is perfect, and agreement across methods increases confidence.

Explanation: This is how research becomes “more believable.”

SAQ 5: Classify standards, guidelines, and principles using authority and generality.

Answer: Standards: high authority, low generality. Guidelines: lower authority, high generality. Principles: abstract, lower authority, high generality.

Explanation: Teachers love asking “which one is mandatory vs recommended?”

C) 1 Case Study (with answer + explanation)

Case Study: Yango “Apply Voucher + Payment + Scheduling” usability issue

You’re observing 8 students doing the class tasks: book a courier pickup **tomorrow at 10am**, book a truck **tomorrow at 5pm**, buy groceries and apply **Rs 100 voucher**, pay via **Easypaisa/JazzCash**, and enable **traffic display**.

4 out of 8 users fail to apply the voucher even though it exists. 3 users accidentally schedule the courier for today instead of tomorrow. Most users say: “I didn’t see where voucher is” and “I thought time was set correctly.”

Task:

1. Identify **at least 3 Nielsen heuristics** being violated.
2. Propose **specific UI fixes** (not generic “make it better”).
3. Suggest **triangulation**: name 2 additional methods to confirm your findings.

Answer (model solution):

1. Likely violated heuristics:
 - **Visibility of system status**: users can’t clearly see whether voucher is applied or what date/time is selected.
 - **Recognition rather than recall**: voucher option and scheduling controls aren’t visible/obvious, so users must “remember” or hunt.
 - **Error prevention**: the system allows easily selecting the wrong date/time without strong checks (e.g., defaulting to today).
(Also possibly **help users recognize/recover from errors** if the system doesn’t clearly warn “pickup time is in the past” or “date is today”).
2. Specific UI fixes:
 - Add a **voucher strip** on checkout: “Voucher: Rs 100 available → Apply” (always visible). After apply, show “Voucher applied: -Rs 100” with an Undo.
 - For scheduling, use a **clear two-step selector**: first date (with “Tomorrow” label), then time; show a final confirmation line: “Pickup: Tue, 10:00 AM”.
 - Add **error prevention checks**: if user selects today + time earlier than current time, show warning and auto-suggest tomorrow.
 - Add **closure** after scheduling: “Courier booked for tomorrow 10:00 AM” screen.
3. Triangulation methods (to confirm):
 - **Analytics/logs**: check drop-off points and how often “apply voucher” is clicked vs completed.
 - **Short survey**: ask many users “Where did you expect voucher option to be?” and “Was pickup date/time clear?”
This aligns with triangulation’s idea of using multiple methods to strengthen confidence.